



DO-IT-YOURSELF: PROJECT

the language code, enabling NLP to recognise the voice spoken in your chosen language and transfer it to ChatGPT to provide an answer in that same language. NLP then processes the answer and converts it

into voice, enabling you to hear the response in your preferred language.

In the code, we set the language code to our desired language; for example, if you want to use Hindi, you set the language code as “hi.” This

way, NLP will recognise whatever you say in Hindi, and ChatGPT will respond in Hindi, with NLP processing the answer as Hindi voice output. Fig. 3 illustrates a code snippet that sets the language code to be used.

Now that our code is ready, you can run it and speak in Hindi or any other language code you have set in the code. VoiceGPT will recognise and respond accordingly using ChatGPT. Fig. 4 showcases VoiceGPT recognising and responding in Hindi, providing both text and voice output.

Multilingual voice chat with ChatGPT

But wait, we promised to make it multilingual, which means it will automatically detect the language and change the language for communication. You only need to run the code once; there is no need to manually change the language code

```
with sr.Microphone() as source:
    r.adjust_for_ambient_noise(source)
    print("Say something!")
    audio = r.listen(source)
    print("Recognizing Now .... ")
    try:
        # using google speech recognition
        #command=str(r.recognize_google(audio))
        text = r.recognize_google(audio, language="hi")
        command=str(text)
        print(command)
    except Exception as e:
        print("Error: " + str(e))
        command=""
    print("Google Speech Recognition thinks you said " + command)
    query=command
    (res, usage) = chatGPT(query)
    print(res)
    tts = gTTS(text=res, lang='hi')
    tts.save("good.mp3")
    pygame.mixer.music.load("good.mp3")
    pygame.mixer.music.play()
    #speak.synth(res)
```

Fig. 3: Code snippet setting the language code to use

Edit Format Run Options Window Help	*Python 3.7.3 Shell*
<pre>t speech_recognition as sr t math t time t serial espeak import espeak t sys t openai t pygame gtts import gtts e.mixer.init() t pyttsx3 _to_use="text-davinci-003" # most capable l_to_use="text-curie-001" l_to_use="text-babbage-001" l_to_use="text-ada-001" # lowest token cost r.Recognizer() ai.api_key="sk-o1GZ7anMRPEVDYaYp2QZT3B1bkFJqer ai.api_key="sk-cUxqAbooiQ30AoyTvm1dT3B1bkFJm1 i.api_key="sk-m6b5oZyPD5RRJ3PR1BhxT3B1bkFJva5(hatGPT(query): response = openai.Completion.create(model=model_to_use, prompt=query, temperature=0, max_tokens=1000) return str.strip(response['choices'][0]['text']) etect_wake_word(): = sr.Recognizer() with sr.Microphone() as source: r.adjust_for_ambient_noise(source) print("Say 'Jarvis' to activate the assistant") audio = r.listen(source) ry:</pre>	<pre>File Edit Shell Debug Options Window Help line_api_resource.py", line 160, in create request_timeout=request_timeout, File "/usr/local/lib/python3.7/dist-packages/openai/api_requestor.py", line 22 7, in request resp, got_stream = self._interpret_response(result, stream) File "/usr/local/lib/python3.7/dist-packages/openai/api_requestor.py", line 62 4, in _interpret_response stream=False, File "/usr/local/lib/python3.7/dist-packages/openai/api_requestor.py", line 68 1, in _interpret_response_line rbody, rcode, resp.data, rheaders, stream_error=stream_error openai.error.InvalidRequestError: This is a chat model and not supported in the v1/completions endpoint. Did you mean to use v1/chat/completions? r >>> ===== RESTART: /home/pi/speech_recognition/examples/hindigpt.py ===== pygame 1.9.4.post1 Hello from the pygame community. https://www.pygame.org/contribute.html LED is ON while button is pressed (Ctrl-C for exit). Say 'Jarvis' to activate the assistant. Jarvis Say something! Recognizing Now क्यांम थयोरि क्या होता है Google Speech Recognition thinks you said क्यांम थयोरि क्या होता है ? क्यांम थयोरि एक प्रकार का डेटाबेस है जो कि डेटा को संग्रहीत करने के लिए उपयोग किया जाता है। यह एक वरिलेख ण और रियरटिंग के लिए उपयोगी है। यह डेटा को संग्रहीत करने के लिए एक सटुक्कर के रूप में काम करता है जो कि डेटा को संग्रहीत करने और वरिलेख करने के लिए आसान बनाता है। Say 'Jarvis' to activate the assistant. Say 'Jarvis' to activate the assistant. Say 'Jarvis' to activate the assistant. Say 'Jarvis' to activate the assistant. Say 'Jarvis' to activate the assistant. Say 'Jarvis' to activate the assistant.</pre>

Fig. 4: VoiceGPT recognising Hindi and talking in Hindi with text as well as voice as output